



Mata Kuliah : Pemrograman Mobile
Program Studi : D4 – Teknik Informatika
Semester : 5

Kelas : 3E TI
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Jobsheet ke : 1

LAPORAN PRAKTIKUM

Praktikum: aplikasi Flutter pertama

Langkah	Keterangan
1.	Membuat dan menjalankan proyek <pre>PS C:\Users\muham> flutter create my_first_app Creating project my_first_app... Resolving dependencies in 'my_first_app'... (2.1s) Downloading packages... Got dependencies in 'my_first_app'. Wrote 131 files. All done! You can find general documentation for Flutter at: https://docs.flutter.dev/ Detailed API documentation is available at: https://api.flutter.dev/ If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev In order to run your application, type: \$ cd my_first_app \$ flutter run Your application code is in my_first_app\lib\main.dart. PS C:\Users\muham> flutter create my_first_app Terminate batch job (Y/N)? ^CError: Unable to determine engine version... PS C:\Users\muham> cd my_first_app</pre> <pre>BUILD FAILED in 34s Running Gradle task 'assembleDebug'... 37.5s Error: Gradle task assembleDebug failed with exit code 1 PS C:\Users\muham\my_first_app> flutter clean Deleting build... 5ms Deleting .dart_tool... 7ms Deleting ephemeral... 8ms Deleting Generated.xcconfig... 6ms Deleting flutter_export_environment.sh... 2ms Deleting ephemeral... 7ms PS C:\Users\muham\my_first_app> flutter run Resolving dependencies... Downloading packages... material_color_utilities 0.13.0 (0.13.1 available) meta 1.18.3 (1.19.0 available) test_api 0.7.12 (0.7.14 available) vector_math 2.4.0 (2.4.2 available) Got dependencies! 4 packages have newer versions incompatible with dependency constraints. Try 'flutter pub outdated' for more information. Launching lib/main.dart on sdk gphone16k x86 64 in debug mode... WARNING: A restricted method on java.lang.System has been called WARNING: java.lang.System::load has been called by net.rubygrapefruit.platform.internal.NativeLibraryLoader in an unnamed module (C:\Program Files\Java\jdk-9.0.4\bin\java.exe) WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module WARNING: Restricted methods will be blocked in a future release unless native access is enabled Checking the license for package NDK (Side by side) 28.2.13676358 in C:\Users\muham\AppData\Local\Android\Sdk\licenses License for package NDK (Side by side) 28.2.13676358 accepted. Preparing "Install NDK (Side by side) 28.2.13676358 v.28.2.13676358". "Install NDK (Side by side) 28.2.13676358 v.28.2.13676358" ready. Installing NDK (Side by side) 28.2.13676358 in C:\Users\muham\AppData\Local\Android\Sdk\ndk\28.2.13676358 "Install NDK (Side by side) 28.2.13676358 v.28.2.13676358" complete. "Install NDK (Side by side) 28.2.13676358 v.28.2.13676358" finished. Checking the license for package Android SDK Platform 36 in C:\Users\muham\AppData\Local\Android\Sdk\licenses License for package Android SDK Platform 36 accepted. Preparing "Install Android SDK Platform 36 (revision 2)". "Install Android SDK Platform 36 (revision 2)" ready. Installing Android SDK Platform 36 in C:\Users\muham\AppData\Local\Android\Sdk\platforms\android-36 "Install Android SDK Platform 36 (revision 2)" complete.</pre>



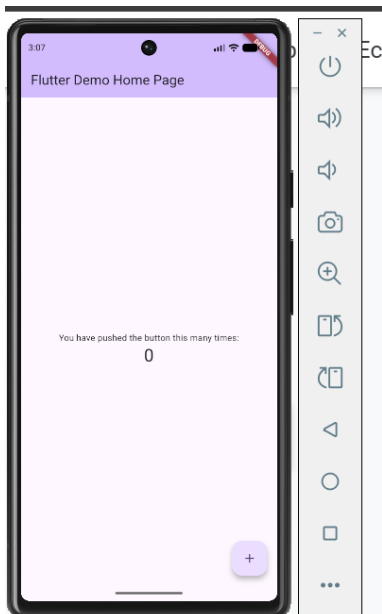
```
"Install Android SDK Platform 36 (revision 22)" complete.
"Install Android SDK Platform 36 (revision 22)" finished.
=====
Checking the license for package CMake 3.22.1 in C:\Users\muham\AppData\Local\Android\Sdk\licenses
License for package CMake 3.22.1 accepted.
Preparing "Install CMake 3.22.1 v.3.22.1".
"Install CMake 3.22.1 v.3.22.1" ready.
Installing CMake 3.22.1 in C:\Users\muham\AppData\Local\Android\Sdk\cmake\3.22.1
"Install CMake 3.22.1 v.3.22.1" complete.
"Install CMake 3.22.1 v.3.22.1" finished.
Running Gradle task 'assembleDebug'...
1467.2s
A Build task failed: flutter-app-debug.apk
Installing build\app\outputs\flutter-apk\app-debug.apk...
28.8s
I/flutterActivityAndFragmentDelegate( 6375): If you are attempting to set --enable-dart-profiling via Intent extras to launch a Flutter component outside of
using the Flutter CLI, note that support for setting engine flags on Android via Intent will soon be dropped; see https://github.com/flutter/flutter/issues/
/180686 for more information on this breaking change. To migrate, set --enable-dart-profiling or any other flags specified via Intent extras on the command
line instead or see https://github.com/flutter/flutter/blob/main/docs/engine/Flutter-Android-Engine-Flags.md for alternative methods.
D/flutterJNI( 6375): Beginning load of flutter...
D/flutterJNI( 6375): flutter (null) was loaded normally!
I/flutter( 6375): [IMPORTANT:Flutter/shell/platform/android/android_context_gl_impeller.cc(104)] Using the Impeller rendering backend (OpenGL).
I/le.my_first_app( 6375): hiddenapi: Accessing hidden method Landroid/view/accessibility/AccessibilityNodeInfo;~>getSourceNodeId()J (runtime_flags=0, domain
=platform, api=unsupported, test-api) from Lio/flutter/view/AccessibilityViewEmbedder$ReflectionAccessors; (domain=app, TargetSdkVersion=36) using reflection
: allowed
I/le.my_first_app( 6375): hiddenapi: Accessing hidden method Landroid/view/accessibility/AccessibilityRecord;~>getSourceNodeId()J (runtime_flags=0, domain=p
latform, api=unsupported) from Lio/flutter/view/AccessibilityViewEmbedder$ReflectionAccessors; (domain=app, TargetSdkVersion=36) using reflection: allowed
I/le.my_first_app( 6375): hiddenapi: Accessing hidden field Landroid/view/accessibility/AccessibilityNodeInfo;~>getChildNodeIds()Landroid/util/LongArray; (run
ime_flags=0, domain=platform, api=unsupported) from Lio/flutter/view/AccessibilityViewEmbedder$ReflectionAccessors; (domain=app, TargetSdkVersion=36) using
reflection: allowed
I/le.my_first_app( 6375): hiddenapi: Accessing hidden method Landroid/util/LongArray;~>get(I)J (runtime_flags=0, domain=platform, api=unsupported) from Lio/
Flutter/view/AccessibilityViewEmbedder$ReflectionAccessors; (domain=app, TargetSdkVersion=36) using reflection: allowed
D/AccessibilityBridge( 6375): Setting not found: accessibility_display_inversion_enabled, using default: 0
D/FlutterRenderer( 6375): Width is zero. 0.0
D/CompatChangeReporter( 6375): Compat change id reported: 377864165; UID 10229; state: ENABLED
W/HWUI ( 6375): Unknown dataspace 0
W/libc ( 6375): Access denied finding property "vendor.mesa.virtgpu.kumquat"
Syncing files to device sdk gphone16k x86 64...
1,782ms

Flutter run key commands.
R Hot reload.
R Hot restart.
h List all available interactive commands.
```

```
W/HWUI ( 6375): Unknown dataspace 0
W/libc ( 6375): Access denied finding property "vendor.mesa.virtgpu.kumquat"
Syncing files to device sdk gphone16k x86 64...
1,782ms

Flutter run key commands.
R Hot reload.
R Hot restart.
h List all available interactive commands.
d Detach (terminate "flutter run" but leave application running).
c Clear the screen
q Quit (terminate the application on the device).

A Dart VM Service on sdk gphone16k x86 64 is available at: http://127.0.0.1:52368/dzHyX94_m0w=/
The Flutter DevTools debugger and profiler on sdk gphone16k x86 64 is available at:
http://127.0.0.1:52368/dzHyX94_m0w=/devtools/?url=http://127.0.0.1:52368/dzHyX94_m0w=/
D/ProfileInstaller( 6375): Installing profile for com.example.my_first_app
I/Choreographer( 6375): Skipped 229 frames! The application may be doing too much work on its main thread.
D/WindowOnBackDispatcher( 6375): setOnBackPressedCallback (unwrapped): android.app.Activity$ExternalSyntheticLambda8@581a41e
D/WindowOnBackDispatcher( 6375): setOnBackPressedCallbackInfo: android.app.Activity$ExternalSyntheticLambda8@581a41e
I/WindowExtensionsImpl( 6375): Initializing Window Extensions, vendor API level=10, activity embedding enabled=true
W/UiContextUtils( 6375): Requested context is a non-UI Context. Creating a UI-Context with display: 0. Context: Context-android.app.Application@b90047, of w
hich baseContext=android.app.ContextImpl@968de22
D/VR[MainActivity]( 6375): WindowInsets changed: 1888x2488 statusBars:[0,0,0,0] navigationBars:[0,0,0,63] mandatorySystemGestures:[0,160,0,84]
D/FlutterRenderer( 6375): Width is zero. 0.0
I/le.my_first_app( 6375): Compiler allocated 5567KB to compile void android.view.ViewGroupImpl.performTraversals(long)
I/Surface( 6375): Creating surface for consumer unnamed-6375-0 with slotExpansion=1 for 64 slots
I/Surface( 6375): Creating surface for consumer VR[MainActivity]H0(BLAST Consumer)0 with slotExpansion=1 for 64 slots
D/FlutterJNI( 6375): Sending viewport metrics to the engine.
I/Surface( 6375): Creating surface for consumer unnamed-6375-1 with slotExpansion=1 for 64 slots
I/Surface( 6375): Creating surface for consumer 2c3472c SurfaceView[com.example.my_first_app/com.example.my_first_app.MainActivity]41(BLAST Consumer)1 with
slotExpansion=1 for 64 slots
I/Choreographer( 6375): Skipped 224 frames! The application may be doing too much work on its main thread.
D/FlutterJNI( 6375): Sending viewport metrics to the engine.
D/WindowLayoutComponentImpl( 6375): Register WindowLayoutInfoListener on Context=com.example.my_first_app.MainActivity@3d5c9fb, of which baseContext=android
.app.ContextImpl@968de22
D/FlutterJNI( 6375): Sending viewport metrics to the engine.
D/InsetsController( 6375): Hide(imc())
I/ImeTracker( 6375): com.example.my_first_app:8cl9d4f5: onCancelled at PHASE_CLIENT_ALREADY_HIDDEN
I/le.my_first_app( 6375): Background concurrent mark compact GC freed 1845KB AllocSpace bytes, 3(96KB) LOS objects, 49% free, 2756KB/5512KB, paused 586us/76
.749ms total 166.958ms
```





2. Mengubah UI default

Buka lib/main.dart, ganti isinya dengan kode berikut, simpan, dan amati hot reload.

Ganti Nama Anda dengan nama sendiri. Ubah ikon atau teks sekali, lalu bandingkan hasil hot reload dan hot restart.

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(title: const Text('Profil Mahasiswa')),
        body: const Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.min,
            children: [
              Icon(Icons.school, size: 72),
              SizedBox(height: 16),
              Text(
                'Muhammad Shabran',
                style: TextStyle(fontSize: 24),
              ),
              Text('Pemrograman Mobile [ ] Minggu 1'),
            ],
          ),
        ),
      ),
    );
  }
}
```

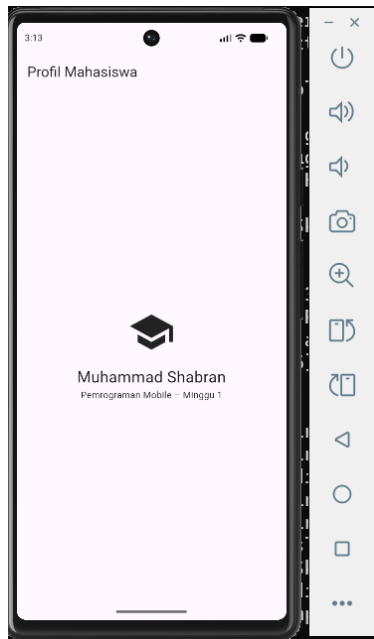
Melakukan Hot reload:

```
Performing hot reload...
Reloaded 1 of 754 libraries in 35,428ms (compile: 3186 ms, reload: 19318 ms, reassemble: 6560 ms).
```

Melakukan Hot Restart:

```
Performing hot restart...
Restarted application in 24,473ms.
```

Tampilan Aplikasi:



Hot Reload (r) berfungsi untuk memperbarui tampilan UI secara langsung tanpa menghapus status (*state*) aplikasi, sehingga perubahan kode pada widget atau teks dapat terlihat secara instan. Sebaliknya, Hot Restart (R) memuat ulang seluruh sistem aplikasi dari fungsi awal `main()`, yang menyebabkan seluruh status data tereset kembali ke kondisi awal seperti saat aplikasi baru dinyalakan.

Mini assignment

Buat aplikasi Profil Mahasiswa berdasarkan praktikum. Tambahkan NIM dan satu informasi tambahan menggunakan widget dasar. Push hasil ke repository portfolio sesuai struktur yang ditentukan. Sertakan screenshot dan penjelasan singkat atas satu kendala setup yang Anda temui.



```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Profil Mahasiswa',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.blue),
        useMaterial3: true,
      ),
      home: const StudentProfilePage(),
      debugShowCheckedModeBanner: false,
    );
  }
}

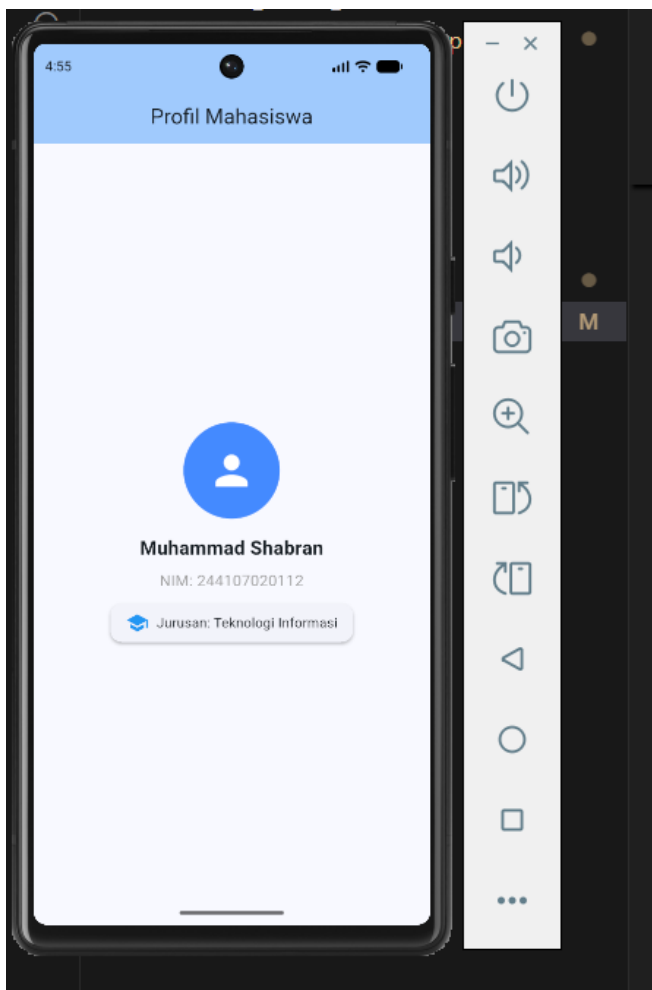
class StudentProfilePage extends StatelessWidget {
  const StudentProfilePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Profil Mahasiswa'),
        backgroundColor: Theme.of(context).colorScheme.inversePrimary,
        centerTitle: true,
      ),
      body: Center(
        child: Padding(
          padding: const EdgeInsets.all(16.0),

```

```
child: Column(
  mainAxisAlignment: MainAxisAlignment.center,
  children: [
    const CircleAvatar(
      radius: 50,
      backgroundColor: Colors.blueAccent,
      child: Icon(
        Icons.person,
        size: 50,
        color: Colors.white,
      ),
    ),
    const SizedBox(height: 16),
    const Text(
      'Muhammad Shabran',
      style: TextStyle(
        fontSize: 20,
        fontWeight: FontWeight.bold,
      ),
    ),
    const SizedBox(height: 8),
    const Text(
      'NIM: 244107020112',
      style: TextStyle(
        fontSize: 16,
        color: Colors.grey,
      ),
    ),
    const SizedBox(height: 8),
    Card(
      elevation: 2,
      child: Padding(

```





1. Kapan native lebih tepat dipilih daripada cross-platform?

Answer:

Native lebih tepat dipilih ketika aplikasi membutuhkan akses penuh dan *day-one support* terhadap fitur hardware khusus (seperti kamera pro, Bluetooth LE, sensor IoT, atau AR/VR), performa komputasi latar belakang berlatensi rendah, serta standar keamanan tingkat tinggi seperti pada aplikasi perbankan. Pendekatan ini juga ideal jika Anda ingin menyajikan UI/UX yang 100% patuh pada panduan desain asli masing-masing OS tanpa kompromi performa akibat lapisan *bridge* atau *rendering engine* pihak ketiga.

2. Bagaimana perubahan state berhubungan dengan widget tree dan UI deklaratif?

Answer:

Dalam paradigma UI deklaratif seperti Flutter, UI merupakan fungsi dari *state* ($UI = f(state)$), di mana visual yang tampil di layar hanyalah cerminan dari data yang ada saat itu. Ketika terjadi perubahan *state* (misalnya via `setState`), Flutter tidak mengubah elemen visual secara langsung melainkan memicu proses *rebuild* pada bagian *widget tree* yang relevan untuk mendeskripsikan ulang konfigurasi UI baru. *Framework* kemudian membandingkan *widget tree* baru tersebut dengan struktur sebelumnya (*reconciliation/diffing*) dan secara efisien hanya memperbarui elemen *render tree* yang mengalami perubahan data di layar.

3. Mengapa commit kecil dengan pesan jelas bermanfaat bagi pekerjaan tim dan portfolio?

Answer :

Commit kecil dengan pesan yang jelas memudahkan tim dalam melakukan *code review*, melacak riwayat perubahan, serta mengisolasi dan membatalkan *bug* secara presisi tanpa mengganggu kode lainnya. Bagi portofolio, kebiasaan ini mencerminkan alur kerja profesional dan struktur berpikir yang terorganisir, sehingga merekruter dapat melihat proses pemecahan masalah Anda secara transparan melalui *commit history* yang rapi.